

Enterprise Learning & Workforce Management Platform

Milestone 4 – Career Readiness, Platform Utilities & Remaining Admin Modules

Resume Builder · Jobs & Internships · Profile & Settings · Notifications · AI Assistant · Support & Help · Remaining Admin Tools

Mentor: Shakti Gopal Krishnan

Team: Kartik · Kavipriya · Harshit · Gayatri

1. Objective

Milestones 1 through 3 built authentication, the core learning loop, and skill-building/competitive features. Milestone 4 closes the loop by turning learning into career readiness, and by rounding out the platform utilities and administrative tooling every production system needs — the pieces that make the platform feel complete rather than just functionally correct.

The main objectives were:

- Let students build and export a job-ready resume from inside the platform.
- Surface relevant job and internship opportunities students can apply to directly.
- Give students a Profile to showcase their work and Settings to control their account.
- Add an in-app Notification centre for platform events.
- Add an AI Assistant chatbox for instant, conversational help.
- Complete the Support & Help ticketing system for issues the AI Assistant can't resolve.
- Finish the Admin Dashboard with revenue analytics, enrollment analytics, complaint management, audit logging, and CSV report export.

2. Resume Builder

The Resume Builder lets a student assemble a professional resume without leaving the platform, using the same profile, course, and certificate data already captured elsewhere in the system.

2.1 Building Flow

```

Student opens Resume Builder
↓
Enter education, skills, projects, experience, achievements
↓
Save → stored as resume data
↓
Preview → rendered in final resume layout
↓
Export → PDF generated (OpenPDF / jsPDF + html2canvas)
  
```

2.2 ATS Score

A live ATS (Applicant Tracking System) score gives the student instant feedback on how well their resume is likely to parse in automated hiring pipelines, alongside a checklist of common resume best practices and ready-to-use templates.

3. Jobs & Internships

The Jobs & Internships section surfaces opportunities relevant to a student's chosen career track.

```

Student searches / filters listings
↓
Opens a listing → role, requirements, location, salary
↓
Apply Now
↓
Application uses existing profile + resume
↓
No separate document upload needed
  
```

Implementation note: in the current build, job listings are managed and stored on the client side (via JobService.js) rather than through a persisted backend table. The admin-side ManageJobs screen curates the same listings. Wiring this to a proper Job entity and JobController with a jobs table is a natural next step beyond this milestone.

4. Profile & Settings

4.1 Profile

Profile is the student's professional snapshot — personal information, skills, social links (LinkedIn, GitHub, portfolio), achievements, and learning statistics such as courses enrolled, certificates earned, and average progress. These statistics are read from the same users, enrollments, and certificates tables already built in Milestones 1 and 2, not duplicated into new storage.

4.2 Settings

Settings covers account-level preferences: notification toggles, password change, and a Light/Dark theme switch. The theme choice is persisted to the browser's localStorage and applied through the same Auth & Theme Context introduced for the public landing page in Milestone 1, so the chosen theme stays consistent whether the student is looking at the public site or the authenticated dashboard.

5. Notifications

An in-app notification centre keeps students informed of platform events without needing email — a certificate being issued, a complaint being resolved, a contest result being posted, and similar events.

```

Platform event occurs (e.g. certificate issued)
↓
Notification record created for the user
↓
GET /api/notifications → shown in notification centre
↓
PUT /{id}/read or PUT /read-all

```

6. AI Assistant

The AI Assistant is a chatbox available across the platform. A student can type any question — about a course, the career roadmap, or a general doubt — and receive an instant, conversational reply, without needing to file a support ticket for a quick question.

```

Student types a question in the chatbox
↓
POST /api/chat
↓
ChatController → GeminiService
↓
Google Gemini API generates a response
↓
Reply streamed back to the chatbox

```

The AI Assistant acts as a fast first line of help, complementing rather than replacing the Support & Help ticketing system — quick questions get an instant answer here, while issues that need a human are raised as a ticket.

7. Support & Help

Support & Help is the platform's built-in helpdesk for issues that genuinely need human attention.

```

Student raises a ticket
(category, subject, description, priority, optional screenshot)
↓
Complaint row created (status = Pending)
↓

```

```

Admin reviews in Complaints Management
↓
Status updates: Pending → In Progress → Resolved / Closed
↓
Optional admin_reply visible to the student
↓
Ticket remains in student's history for follow-up

```

Categories supported are Technical, Billing, Course Content, Account, and General, with priority levels Low, Medium, High, and Urgent, giving administrators enough context to triage without needing to open every ticket individually.

8. Remaining Admin Modules

8.1 Revenue Analytics

Gives administrators visibility into platform revenue: monthly trends, a per-course revenue breakdown, and a transaction history, backed by dedicated payments, transactions, and revenue_reports tables, with CSV export for offline reporting.

8.2 Student Enrollment Analytics

Provides enrollment trend charts and per-student progress views for administrators, backed by a dedicated analytics table separate from the raw enrollments table, so heavier analytical queries don't compete with the live enrollment/progress reads used throughout the student experience.

8.3 Complaints Management

The admin-side counterpart to Support & Help: administrators view incoming tickets, prioritise by urgency, and resolve them by updating status and writing a reply, closing the loop opened by the student in Section 7.

8.4 Audit Logs

Every sensitive administrative action is recorded — user email, action description, IP address, and timestamp — in an audit trail, giving accountability and traceability over what admins have changed on the platform.

8.5 Report Export

CSV export endpoints for users and courses let administrators pull platform data into spreadsheets for offline reporting or record-keeping outside the application.

9. Core Backend Classes

NotificationController.java

Serves and updates the read/unread state of in-app notifications for the current user.

ChatController.java / GeminiService.java

Handles chatbox requests from the frontend and forwards them to the Google Gemini API, returning the generated reply.

ComplaintController.java

Handles ticket creation on the student side and status/reply management on the admin side.

RevenueAnalyticsController.java

Aggregates payment and transaction data into the revenue dashboards and reports.

StudentEnrollmentAnalyticsController.java

Serves enrollment trend and per-student progress data for the admin analytics views.

AuditLogController.java

Records and retrieves the administrative action log.

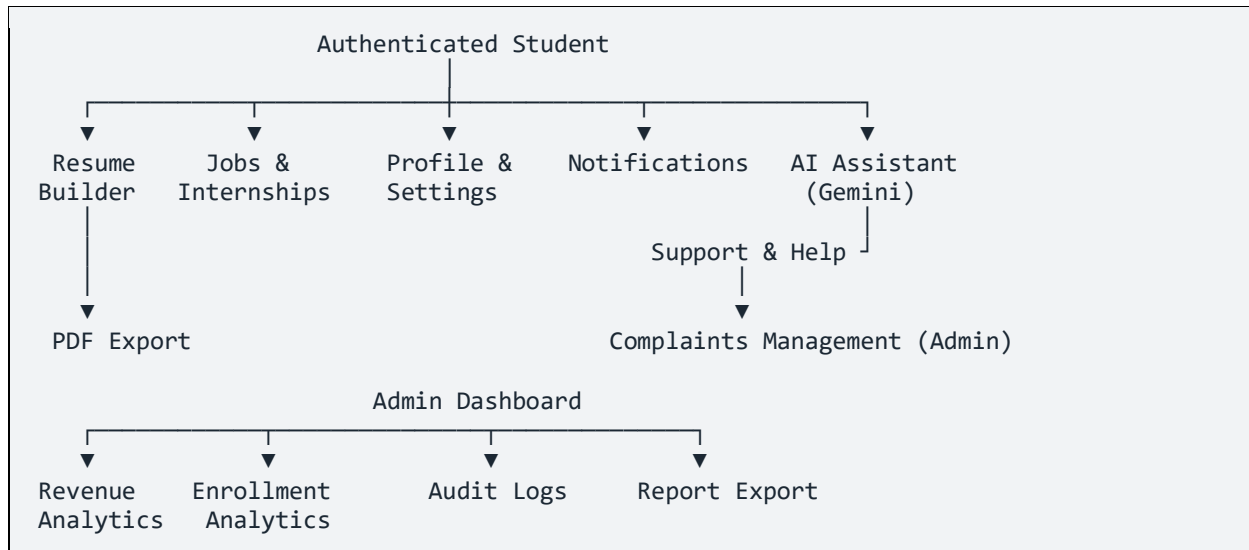
ReportExportController.java

Generates CSV exports of users and courses on request.

10. Database Design for This Milestone

- notifications — recipient, message, read/unread state, timestamp
- complaints — category, subject, description, screenshot_url, priority, status, admin_reply
- payments, transactions, revenue_reports — revenue analytics
- student_enrollments (analytics) — enrollment trend data separate from the live enrollments table
- audit_log — user_email, action, ip_address, created_at
- users table extended — bio, address, linkedin_url, github_url, portfolio_url used by Profile

11. Complete Milestone 4 Architecture



12. Conclusion

Milestone 4 completes the Enterprise Learning & Workforce Management Platform end to end. Students now have a full path from learning to job readiness — building a resume, discovering relevant jobs and internships, presenting a polished profile, getting instant AI-assisted help, and falling back on a proper support system when needed — while administrators have the revenue visibility, enrollment insight, complaint-handling tools, audit trail, and reporting needed to run the platform responsibly.

Across all four milestones, the platform progressed from a secure entry point (Milestone 1), through a working learning-and-certification loop with admin foundations (Milestone 2), into deeper progress tracking, practice, and competition (Milestone 3), and finally into full career readiness and operational completeness (Milestone 4) — delivering the complete Enterprise Learning Platform with Skill and Career Guidance System.